PRANAV SHARMA

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SUMMARY

Final-year Computer Science student at Thapar Institute of Engineering and Technology specializing in applied machine learning and computer vision. Fine-tuned YOLOv8 to 0.87 mAP50 for blood-cell detection and built the ML models (Random Forest, XGBoost, Naive Bayes) behind a hybrid rockfall early-warning system that won 3rd Place & Innovation Award at Agentic AI Hackathon 2025 (Ulster University, UK). Deploy models and full-stack apps using FastAPI, Flask, Next.js, and Docker.

EDUCATION

B.Tech in Computer Science, Thapar Institute of Engineering and Technology July 2023 – 2027 (Expected)
CGPA: 7.0 / 10.0 (through 6th semester)

St. D.R. Public Senior Secondary School, Gagret 2021 – 2023
Class XII (PCM, 2023): 90.8% | Class X (2021): 94.14%

SKILLS

Languages: Python, C++, C, Java, TypeScript, SQL
ML & Deep Learning: YOLOv8 (transfer learning), Scikit-learn (Logistic Regression, Random Forest), XGBoost, Naive Bayes, OpenCV, NLP, TensorFlow/Keras
Data & Evaluation: NumPy, Pandas, Matplotlib, Feature Engineering, Model Evaluation (mAP, accuracy, precision/recall), TF-IDF
Backend & Databases: FastAPI, Flask, Next.js, REST APIs, PostgreSQL, SQLite
Deployment & Tools: Docker, Git/GitHub, Render, Vercel, Jupyter/Colab, Roboflow
Core CS Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, SDLC & Agile Basics

PROJECTS

Rockfall Prediction & Early Warning System for Open-Pit Mines GitHub | Oct 2025
Agentic AI Hackathon 2025, Ulster University (UK) | *3rd Place & Innovation Award* | *Team CoreSight, 6 members*

- Built the ML models (Random Forest, XGBoost, Naive Bayes) for a hybrid rockfall risk-prediction system, fused with Fuzzy AHP to output clear, interpretable risk levels for mine zones.
- The team-built system serves real-time predictions and alerts via a FastAPI backend to a React + Mapbox dashboard showing mine zones, rainfall trends, live risk alerts, and a risk breakdown (geological, environmental, seismic, human impact) with recommended actions.

Tools: *Python, Random Forest, XGBoost, Naive Bayes, Fuzzy AHP, FastAPI, React, Mapbox*

Blood Cell Detection & Cellscope Web App Model | App

- Fine-tuned COCO-pretrained YOLOv8n on the BCCD dataset (frozen first 10 layers, 30 epochs, 640×640) to detect RBCs, WBCs, and platelets.
- Reached 0.87 mAP50 / 0.60 mAP50-95 (precision 0.79, recall 0.88) on a 36-image held-out test set; per-class mAP50: WBC 0.97, RBC 0.87, platelets 0.78.
- Built Cellscope, a FastAPI inference API with a web UI that returns annotated images, automatic RBC/WBC/platelet counts, and CSV export.

Tools: *Python, Ultralytics YOLOv8, FastAPI, HTML/JS, Vercel, Google Colab, Roboflow*

PictoSpeak AI: OCR-to-Speech Web App GitHub | Live Demo

- Built an OCR-to-speech pipeline (OpenCV preprocessing, Tesseract OCR with editable output, gTTS) that converts uploaded or camera-captured text into English/Hindi audio (playable or MP3 download).
- Shipped as a multi-user Flask app with email-OTP signup, per-user history in SQLite, Docker containerization, and deployment on Render.

Tools: *Python, Flask, OpenCV, Tesseract OCR, gTTS, SQLite, Docker, Brevo API*

Sustainable Harvest Solutions (SHS): Full-Stack Farming App GitHub | Live Demo

- Built a full-stack Next.js app that matches crop breeds to climate zones for farmers, with fertilizer/herbicide cost lookups, market-price comparison, and an English/Hindi UI toggle.
- Implemented a PostgreSQL schema with functions, procedures, cursors, and triggers; secured access with bcrypt-hashed passwords, signed HTTP-only cookie sessions, and email-OTP registration; deployed on Vercel.

Tools: *Next.js, TypeScript, PostgreSQL, SQL, bcrypt, SMTP (email OTP), Vercel*

CERTIFICATIONS

- Building AI-Based Cybersecurity Pipelines, NVIDIA (Nov 2025)
- The Complete Python Developer, Udemy (Sept 2026)